RETAIL CUSTOMER INTELLIGENCE - END-TO-END DATA SCIENCE

Customer & Demand Intelligence

Raw transactions -> cleaning -> star-schema warehouse -> 4 ML analyses -> profit-value model.
Business leads; technology serves.

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PROBLEM & OBJECTIVE

Why this project exists

The problem

- Every shopper treated the same
- Customers leave before anyone notices
- Promotions fired at everyone - discounting people who'd buy anyway
- Stock set by gut
- A chunk of revenue from shoppers we can't identify

The objective

- Grow gross-margin profit by doing four things well:
- Know our customers; keep the valuable ones;
- grow the rest; stock the right products.
- Put a money value on each opportunity to rank investment.
- Prove every recommendation before spending at scale.

WHY THIS APPROACH

The answer is already in the data.

Every problem above is one gap - we don't truly know our customers. So the fix isn't more discounts or more data; it's turning the transactions we already collect into four decisions, each tied to profit. The method follows from that: business-led models, honest baselines, and a controlled experiment behind every play - so we ship what's proven, not what's merely plausible.

Flying blind -> it's a customer-knowledge gap -> segment, predict churn, recommend, forecast

THE FOUNDATION

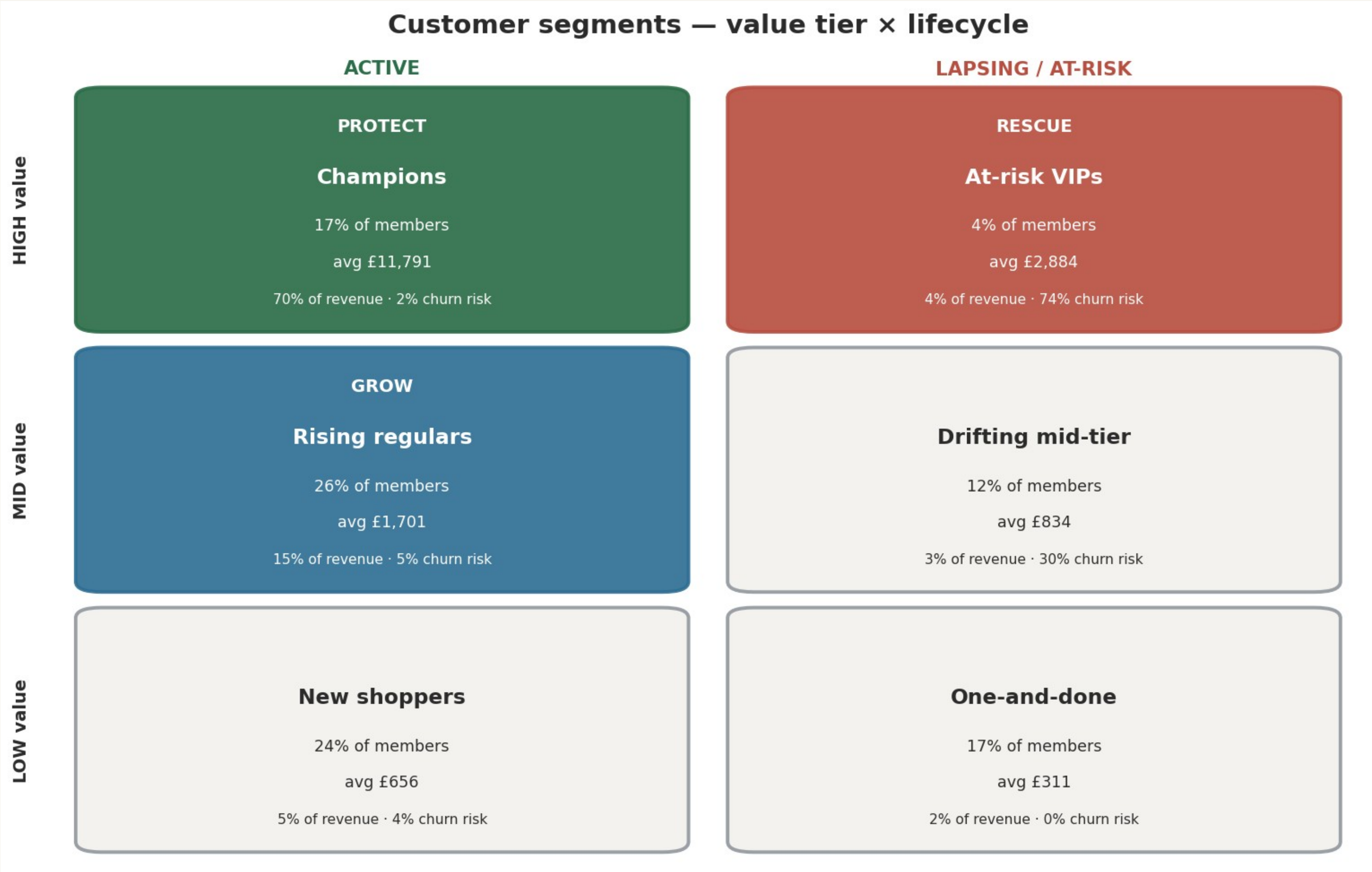
From 1.07M messy rows to a trustworthy warehouse

94%	5 tables	£19.64M	12
of 1.07M rows kept (audited)	star schema (fact + 4 dims)	revenue reconciled, 0 unmapped	LLM-built product categories

- Visible, reusable cleaning (duplicates, cancellations, non-positive, service codes); a modelling
- cleanse that keeps guests labelled Member/Non-Member; a feature foundation with the right
- source-of-truth split (customer models on Members; demand & basket guest-inclusive); and an
- LLM that created the product-category dimension the raw data lacked.

Six behavioural segments (RFMTC + K-Means)

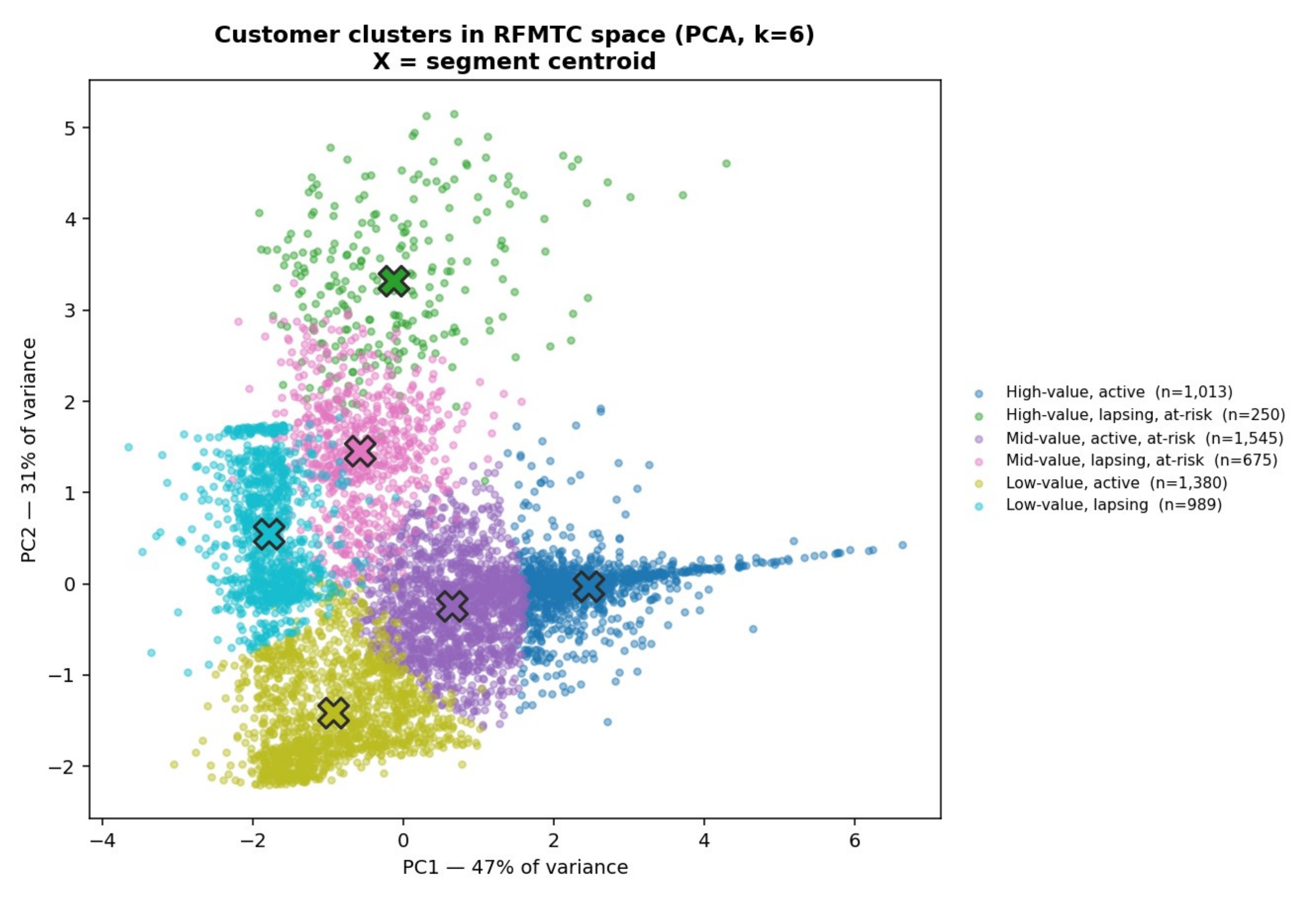
- Features: Recency, Frequency, Monetary, Tenure + a churn-probability term (RFMTC).
- k=6 chosen on elbow + silhouette + business capacity.
- 17% of members drive 70% of revenue.
- Prioritised, one job each: Protect, Rescue, Grow.



Six groups by value tier x lifecycle; priority segments colour-coded.

Clusters in RFMTC space (PCA, 78% variance)

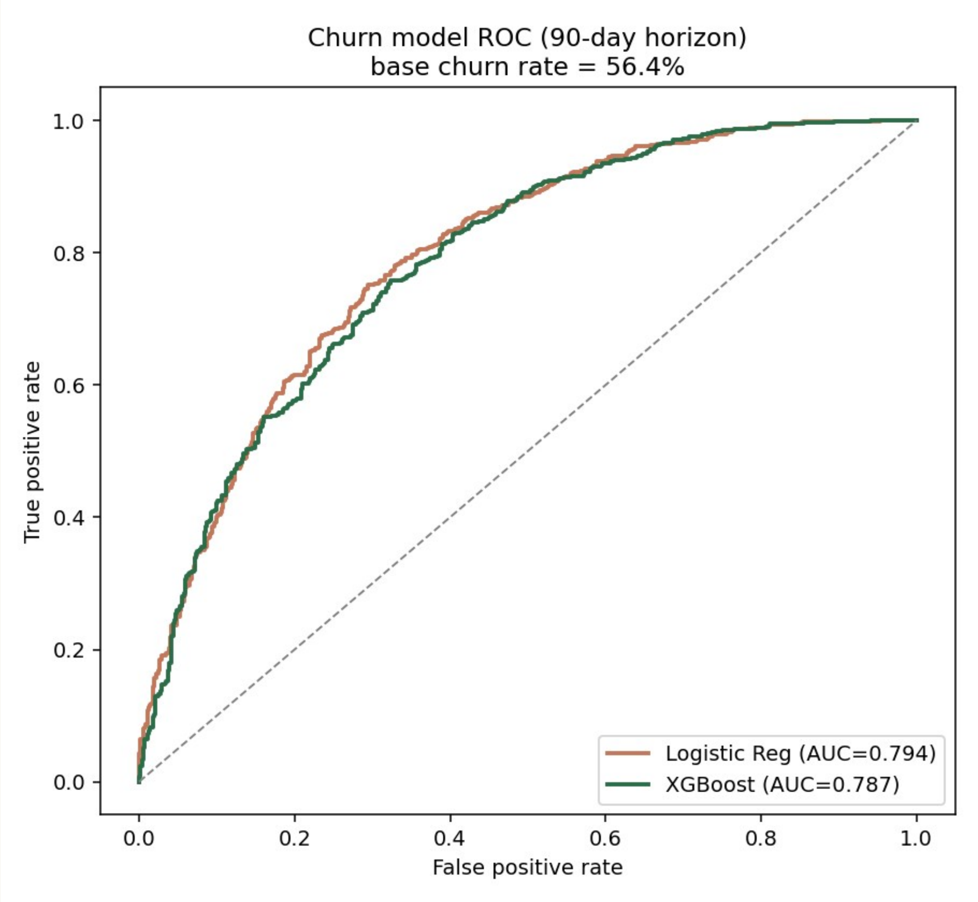
- PCA projection of the standardised RFMTC features, coloured by segment.
- Centroids marked; groups separate along a value-and-activity gradient.
- Honest: silhouette ~0.36 - real groups, but a continuum, not islands.



PCA = flatten the 5 RFMTC features onto a 2-D map so groups can be seen by eye; these two axes keep 78% of the real differences between customers.

Early-warning model - ranks who'll leave (AUC 0.79)

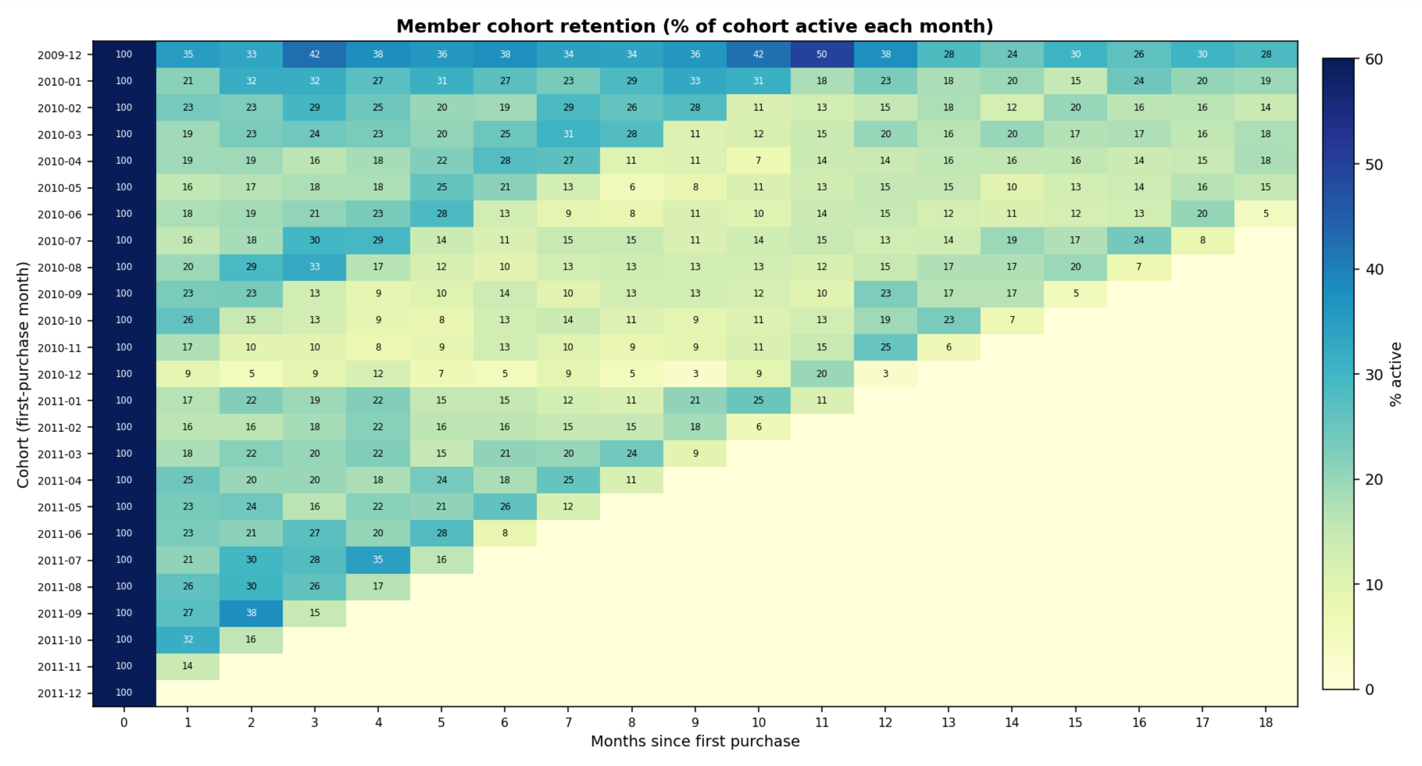
- Strict time split: features before cutoff, label = no purchase in next 90 days.
- AUC 0.79 = ranks a real leaver above a non-leaver ~4 times in 5 (0.5 = coin-flip).
- Logistic ~ XGBoost - signal is mostly linear; reported honestly.
- Top driver: recency vs the customer's normal rhythm. Feeds the rescue list.



ROC: both models ~0.79 vs the 0.5 chance line.

Cohort retention - the first-repeat cliff

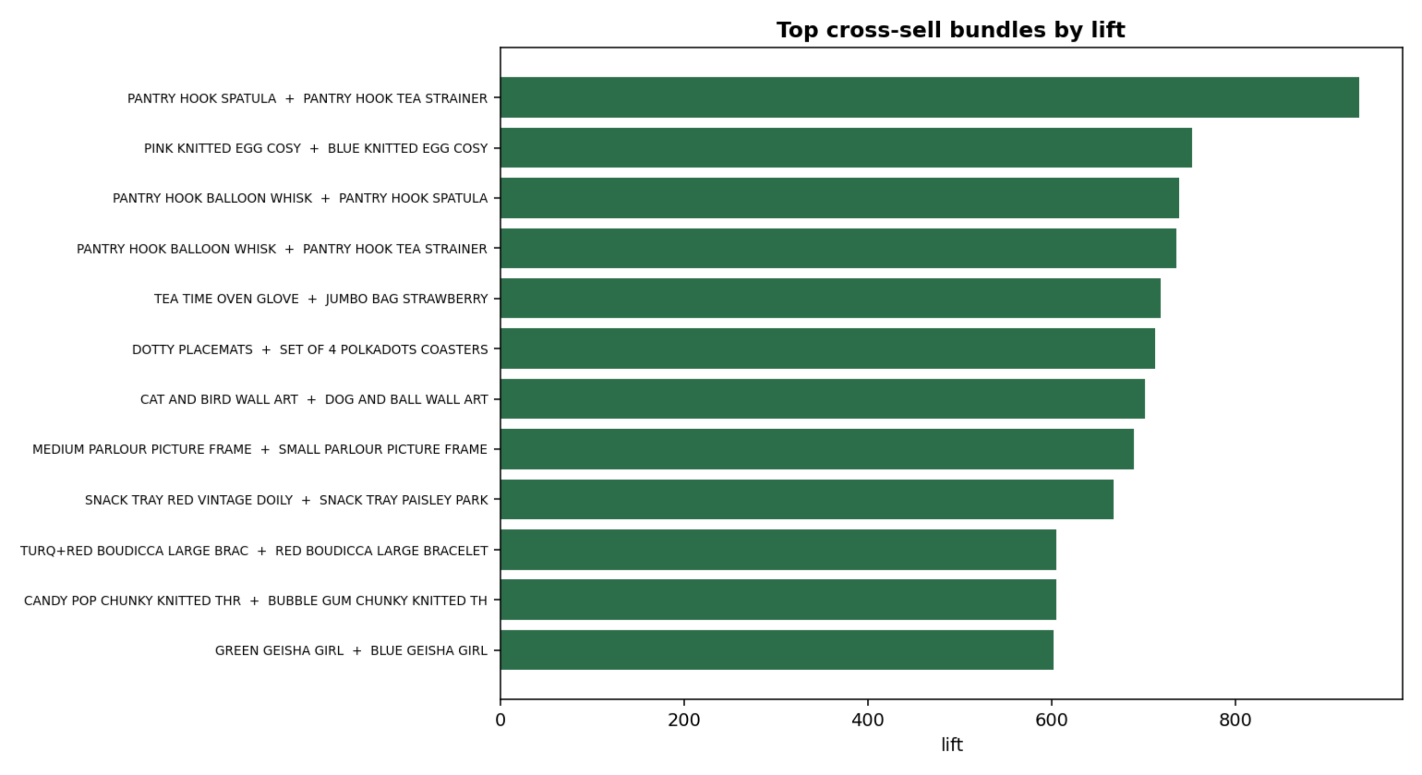
- ~20% return in month 1, settling to ~14% by month 6.
- The leak is the SECOND purchase -> fix with onboarding, not loyalty tiers.
- Cohort curves become the retention KPI to test every initiative.
- Decay rate feeds lifetime value and the value model.



Curves are cohort averages; a few early cohorts tick UP near month 11-12 as seasonal (Christmas) buyers reactivate - retention isn't strictly monotonic in seasonal retail.

Recommender - right product ~1 in 4 (~100x chance)

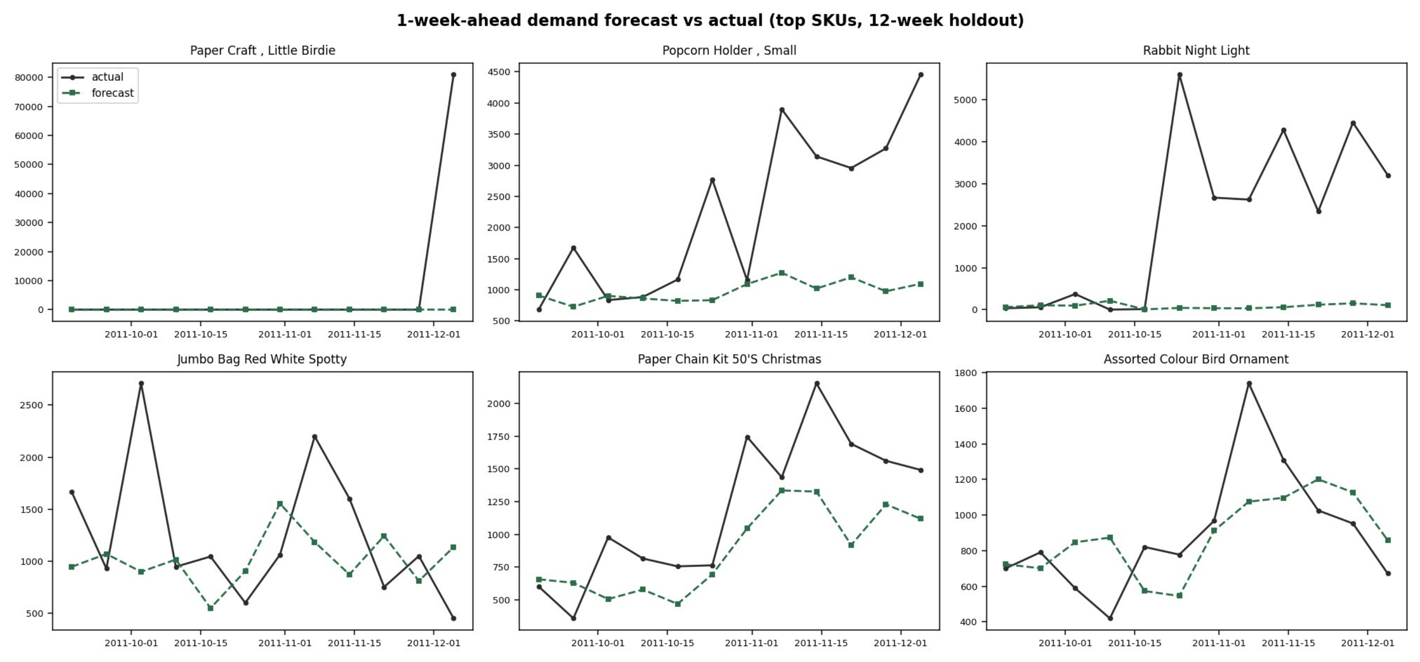
- Item-based collaborative filtering (cosine on the basket x item matrix).
- Leave-one-out hit-rate@10 = 24.6% vs 0.24% random.
- Affinity rules feed bundle ideas (with merchandiser curation).
- Dead-stock paired with high-affinity anchors to clear inventory.



Top cross-sell bundles by lift.

Demand forecast -> profit-aware stocking

- Global XGBoost (log target) vs a 4-week moving-average baseline.
- Honest finding: baseline wins aggregate; XGBoost wins 61% of SKUs -> HYBRID.
- Real-unit error: MAE ~220 units; and it UNDER-forecasts ~126 units/wk -
- left uncorrected that causes stock-outs, so the newsvendor rule adds the buffer.

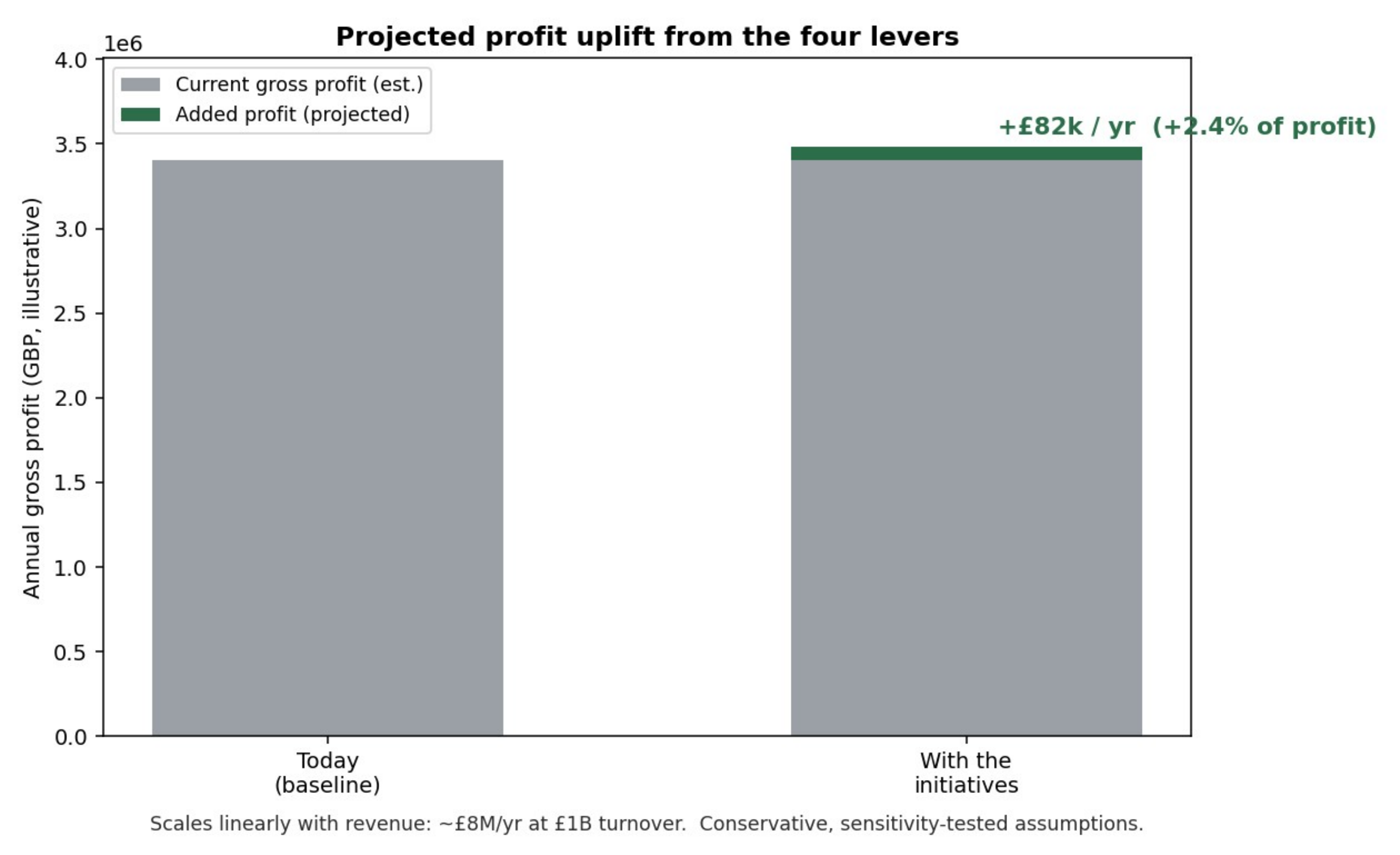


1-week-ahead forecast vs actual, top SKUs (12-week holdout).

THE VALUE

Levers priced in money - and a tangible campaign view

- ~£82k/yr incremental profit (+2.4% of profit)
- Retention is the biggest lever; ranges £41k-123k (uplift), £57k-106k (margin).
- Rescue campaign: ~30 VIPs won back -> ~£43k revenue / ~£15k margin -> net ~£11k, ~3x return.
- Every play ships with a holdout control to prove incremental lift.



Projected uplift; the same % scales with revenue.

How it's built - and what it can't claim

- **Stack**
  - Python, pandas, scikit-learn, XGBoost, lifetimes (BG/NBD)
  - scipy, matplotlib, parquet star schema, reproducible modules
  - LLM-assisted category enrichment (GenAI); leakage-safe splits
- **Evaluation**
  - AUC for ranking; WAPE + MAE/bias (real units) for forecasts
  - Always benchmarked vs a simple baseline; holdout controls

Honest caveats (stated, not hidden)

- Public UK proxy - methods transfer, product mix does not.
- No COGS -> margin is an explicit, sensitivity-tested assumption.
- Lost sales unobservable -> availability lever is assumption-heavy.
- Small cohorts (rescue n=250) -> directional, expand to gain power.
- Simple beat ML on forecast aggregate -> hybrid, not a trophy model.

From data to decisions

Clean foundation -> warehouse -> segmentation, churn, cross-sell, forecasting -> profit value.

Business-first framing, leakage-safe modelling, honest metrics and baselines.

Code, methodology, and limitations: README.md and docs/.